Does a surplus-production ladder improve forecasts of Northern cod? A scored test on NAFO 2J3KL

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Highlights

- A five-module structural ladder is scored against two naive baselines
- No structural model beats last-value persistence on the primary out-of-sample score
- The 1991–1995 collapse is missed by every model under both catch treatments
- No module is retained, scoped to the estimator and series tested
- Stall reconstructions and the ladder share the constant-productivity failure mode

Abstract

Five surplus-production modules and two naive baselines are scored on Northern cod (*Gadus morhua*), NAFO 2J3KL, under a retention rule fixed before scoring: a module is kept only if it lowers primary RMSE against the next-simpler module and against persistence. The predictand is the NCAM M-shift SSB series (DFO, 2016, Table A2), 1983–2015, LRP 884.6 kt, catch entering as a coarse regime and as annual landings.

No module is retained on these two unpooled series. One-year rolling RMSE is 98 kt for persistence against 115–206 kt for the ladder, five-year 265 kt against 289–488 kt. Every model misses the 1991–1995 collapse (694–819 kt, against 670 and 688 kt for the baselines): constant productivity with a 1992 catch drop cannot reproduce the crash. Neither an AR residual nor a one-year delay repairs it (819 kt each); the delay raises one-year RMSE to 196 from 135 kt. Added structure is not uniformly harmful — the autoregressive module reads below the stock-flow module in all six rolling comparisons — but none closes the gap. A second, unpooled specification (xteNCAM, 1954–2024, LRP 276 kt) gives the same outcome (origin-matched persistence 84 kt against 120 kt for M1 at $h = 1$).

The result is scoped to the estimators and series tested. On Specification A the margins are a point-rule ranking within bootstrap noise, intervals excluding zero only on Specification B. The predictand is retrospectively reconstructed rather than the vintages at each origin, and catch is supplied along the horizon: a conditional hindcast, not an operational forecast.

Keywords: northern cod; biomass forecasting; surplus production; forecast evaluation; stock assessment; prediction skill; model identifiability

1. Introduction

Fisheries assessment builds structure by default. State-space age-structured assessments, surplus-production models, and ecosystem-linked extensions each add parameters, state variables, or couplings in the hope of better advice. On Northern cod, that structure does not pay for itself: across a five-module surplus-production ladder scored on two unpooled assessment specifications, no module forecasts spawning-stock biomass more accurately than last-value persistence. The result is scoped to the estimators and series tested, and the exercise is a conditional hindcast rather than an operational forecast test. Whether that structure improves out-of-sample forecasts — as opposed to in-sample fit — is a separate, testable question, and on it the burden of proof sits with the added structure.

That burden is now formalised in assessment practice. Comparison against a naive baseline has become a standard acceptance diagnostic: hindcast cross-validation scores a model's out-of-sample predictions with the mean absolute scaled error (MASE) of Hyndman and Koehler (2006), which divides prediction error by the error of a random-walk forecast, so that a value above one means the model predicts less accurately than assuming no change (Kell et al., 2016; Kell et al., 2021). The diagnostic cookbook of Carvalho et al. (2021) lists prediction skill so measured as one of four criteria for accepting an integrated assessment, alongside convergence, fit, and retrospective consistency, and the good-practice guidance for surplus-production models adopts the same test (Kokkalis et al., 2024). The reference forecast in that literature is therefore already the one used here.

Two features of that practice leave the present question open. First, MASE is almost always computed on the *abundance index* a model fits, not on the estimated stock state that advice is written about, so a model can pass on the index it was tuned to while its biomass trajectory remains untested. Second, the diagnostic is applied to certify a single accepted assessment rather than to rank a graded sequence of structural additions against each other and against the naive rule at once. What is missing is a direct, scored comparison in which the predictand is the assessment's own biomass output, each increment of structure has a declared comparator, and the naive rule is a competitor rather than a scaling constant. That is the comparison reported here.

Northern cod (*Gadus morhua*) in NAFO divisions 2J3KL is the canonical demanding case. The late-1980s and early-1990s collapse — a spawning stock falling from a high, weakly trending phase to a few percent of its 1980s mean — occurred under an assessment and management regime whose failures were dissected in the canonical post-mortems (Hutchings and Myers, 1994; Walters and Maguire, 1996). The non-recovery that followed the July 1992 moratorium has kept the depensation question open for three decades (Shelton and Healey, 1999). A partial rebuilding was documented in the 2010s (Rose and Rowe, 2015), but surplus production has since stalled, with some years being negative (Rose, 2026).

The measured quantity throughout is the condition of the productive stock — the spawning-stock biomass whose persistence underwrites future yield. This is distinct from the catch taken from it. A fishery can return a high extracted yield while the stock that produces it declines, and it is the latter that the forecast here must track. Any model family proposed for this stock must therefore first clear a minimal bar: it must at least beat the forecast that nothing changes. One-dimensional autonomous surplus-production maps carry an a priori structural bar that this stock confronts directly: under constant catch such a map has a repelling lower equilibrium below which no path can recover, and the observed path falls below it. Proposition 4.1 states the condition and the fitted values. The scored test measures how severely that bar penalizes out-of-sample error rather than whether the bar exists.

This evaluation follows a stated retention rule, coded before the first scoring pass and applied unchanged (Section 4 records its protocol status): additional model structure is retained only when it reduces primary RMSE relative to the next-simpler model and relative to persistence (§2.3). Early-warning and intervention-selection criteria are secondary objectives; neither is operationalised here (the Brier score is reported as a secondary diagnostic and never changes a retention verdict, and the moratorium is deliberately not evaluated). The instrument is a forward-ordered ladder of surplus-production models (not a strict nesting for M2 and M4) evaluated against two naive baselines. It runs output-only, stock-and-flow, residual, then delay;

prey- and index-informed observation variants are evaluated separately in Section 3.4 and are not part of the ladder. A module is kept only if it improves the stated score.

The paper asks whether stock-flow, residual, delay, or prey-informed modules reduce SSB forecast error relative to last-value persistence and to the autonomous surplus-production model. The scope is deliberately narrow: this paper does not estimate an Allee threshold, identify the cause of the 1990s collapse, or evaluate whether the 1992 moratorium was adequate.

The stock was selected because of its data availability (a full state-space assessment series), its policy relevance, and the sharpness of its collapse-and-stall trajectory. The same scored design is applied to a groundwater system, the Edwards Aquifer, Texas, in Abaee (2026a). The two systems' series are not pooled, and no retention outcome is transferred between them.

The remainder of the article is organized as follows. Section 2 states the data, the two assessment specifications, the model ladder, and the evaluation design. Section 3 reports the results: the primary specification (3.1), annual landings and the survey-start variant (3.2), the alternative assessment specification (3.3), prey-informed productivity (3.4), the Diebold–Mariano and block-bootstrap uncertainty analysis (3.5), and the fitted parameters (3.6). Section 4 discusses the findings, the identification limits, the reconstruction-level corroboration, and the protocol status. Section 5 concludes.

2. Material and methods

2.1 Data and specifications

The specification tables type each field as follows: D, data; E, empirical construct; M, model; N, normative threshold.

Table 1. Primary specification — Specification A (the 1983–2015 NCAM M-shift SSB series of DFO, 2016, Table A2; the state-space assessment model of Cadigan, 2016).

Field	Contents	Type
System	Northern cod, NAFO 2J3KL, as represented by NCAM M-shift SSB	D
Interest	Continuity of a spawning stock on the 2016 precautionary-approach cautious or healthy side of the 2016 LRP	N
Domain	Stock area in DFO (2016) Figure 1; calendar years 1983–2015	D
Safe set	$S_t \geq$ LRP = 884.6 kt (1983–1989 mean of Table A2)	N
Disturbance	Unspecified productivity shocks; not a fitted $M(t)$	M
Theoretical catch	Any $C_t \geq 0$	M
Implementable catch	Pre-1992 directed fishery; post-2 July 1992 moratorium and low inshore removals	E
Horizon	Hindcast 1983–2015; two fixed test windows and rolling origin	D
Food web	Capelin excluded from the primary pass; tested as a variant in Section 3.4	M
Norm	2010/2016 DFO precautionary-approach LRP (1980s mean SSB), not the 2023 40% B_{MSY} LRP	N

Table 1 note: the safe-set and LRP fields type the reference frame of the evaluation; the primary RMSE score never uses them (only the secondary Brier indicator score does).

Table A2 also reports fishing mortality F and natural mortality M. They are joint outputs with SSB and are not used as exogenous inputs.

Regular et al. (2025) extend the assessment to 1954, revise the LRP to 276 kt (95% interval 180–423 kt; 40% of B_{MSY}), and estimate 2024 SSB at 342 kt. That is a second specification — Specification B (the 1954–2024 extended xteNCAM series; Section 3.3). The two objects differ in four specification fields: the state equation (xteNCAM versus NCAM M-shift; start year 1954 versus 1983), the reference threshold (276 kt versus 884.6 kt), the treatment of catch, and the horizon. The two SSB columns are not mixed. The reconstructions are scored separately because they differ in estimated SSB trajectory, assessment formulation, temporal coverage and catch specification; the reference threshold enters only the secondary classification diagnostic and not the primary RMSE comparison. No retention outcome is carried from one specification to the other.

2.2 Forecast models

Northern cod surplus production is represented as a discrete-time map, with S in kt and C in kt yr^{-1}.

Definition 2.1 (Surplus-production map). The autonomous surplus-production map with multiplicative depensation is

$$S_{t+1} = \big[S_t + g(S_t) - C_t + \varepsilon_t\big]_+, \qquad g(S_t) = rS_t\big(1 - S_t/K\big)\, a(S_t),$$
$$a(S_t) = \begin{cases} 1 & \text{(Schaefer; no Allee term declared),} \\ \dfrac{S_t - \mathfrak{s}}{K - \mathfrak{s}} & \text{(depensation term active, } \mathfrak{s} > 0\text{).} \end{cases}$$

Two conventions fix how this map is used. The disturbance ε_t is a regression error at estimation only: parameters are fitted to the one-step increments with ε_t as the residual, and all forecasts reported here set $\varepsilon_t = 0$, so every forecast trajectory is a deterministic plug-in path rather than a multi-step conditional mean, except in M3 and M4, which carry the fitted residual forward as a state. In those two modules the training residuals are $e_u = \Delta S_u - [g(S_u) - C_u]$ over the training transitions, formed before any clipping. The coefficient $\hat{\phi}$ is the no-intercept lag-one least-squares estimate $\sum_u e_u e_{u-1} / \sum_u e_{u-1}^2$; it is set to zero when the denominator is not positive or the window supplies fewer than four residuals, and is clipped to $[-0.95, 0.95]$. The residual carried into the forecast is the last training residual e_{last}, and the projected residual at step $k \geq 1$ is $\hat{\phi}^k e_{\mathrm{last}}$, added inside the update before the state is clipped to $[10^{-3}, 10^6]$ kt. Catch is dated so that C_t is the catch taken during the interval from t to $t+1$; a forecast issued at origin t therefore uses the recorded catch for the year it is forecasting. Under the coarse regime this places the 1992 drop in the step from 1992 to 1993. For multi-step forecasts the catch path is supplied for the whole horizon, so the exercise is a conditional hindcast with respect to catch, not an operational forecast; Section 4 returns to this.

The factor $a(S_t)$ is a multiplicative depensation term, not a parameter switch. Setting $\mathfrak{s} = 0$ gives $a(S_t) = S_t/K$ (a cubic modification of the logistic surplus), not the Schaefer law. The Schaefer model is the separate member of the family in which the factor is replaced by 1; Abaee (2026b) makes the same distinction.

Section SI-2 gives the statement and proof.

The model ladder is given in Table 2.

The ladder is the forward-ordered set of seven entries {persist, mean, M1, M1b, M2, M3, M4} of Table 2, ordered by structural complexity (autonomous surplus map, Allee extension, stock-flow, residual, delay): two naive baselines (persist, mean) and five structural modules.

Where this article refers to a five-module ladder evaluated against two naive baselines, it means this set. It is not a strict nesting for M2 and M4.

Table 2. Model ladder.

ID	Class	Free on the training window	Frozen into the test window
persist	baseline	—	$\hat{S}_{t+h} = S_t$
mean	baseline	training mean	$\hat{S}_{t+h} = \bar{S}_{\text{train}}$
M1	autonomous	r, K; C = training-mean catch (plugged, not estimated)	same C
M1b	autonomous with Allee	$r, K, \mathfrak{s}$; C = training-mean catch (plugged, not estimated)	same C
M2	stock-flow	r, K; C_t prescribed	prescribed C_t
M3	residual	M2 + AR(1) residual	ϕ persisted
M4	delay	r, K and ϕ refitted exactly as for M3	forecast starts from S_{t-1}

The coarse catch regime, taken from DFO (2016) prose (the precautionary-approach framework of DFO, 2009), is $C_t = 240$ kt for $t \leq 1991$, $C_t = 120$ kt for $t = 1992$, and $C_t = 5$ kt for $t \geq 1993$. Year-by-year landings (Schijns et al., 2021, Table 1) replace that regime in Section 3.2. Regular et al. (2025) Table 1 matches Schijns exactly on 1983–1993 (11 years, maximum absolute difference 0 t); STATLANT matches Schijns on 1983–1993, so the STATLANT-versus-Schijns sensitivity is closed on 1983–1993 (they are the same column there). The Specification A collapse test window runs to 1995, two years beyond the verified match; the residual exposure is immaterial, since Schijns records 1.31 kt in 1994 and 0.41 kt in 1995 against a collapse-window ΔS of hundreds of kilotonnes. A 1956 discrepancy between those sources (236,210 t versus 263,210 t) lies outside the 1983–2015 Specification A range; Specification B begins in 1954 and so contains 1956, but it draws its catch column from the xteNCAM reconstruction rather than from either of these two sources, so the discrepancy is unused on both specifications. The coarseness of the regime series is a limitation.

Parameters are estimated by one-step least squares on the training window only. Rolling-origin training windows expand rather than slide: each origin refits on all history up to that origin, subject to the minimum length. A window of m annual states supplies $m - 1$ one-step transitions, and it is the transition count that sets the effective regression sample. Bounds: $r \in (0.001, 2]$, and K constrained above the largest state the one-step regression conditions on. The estimation code optimises K on $[\max_{\text{train}} S + 10, 5000]$ kt, where $\max_{\text{train}} S$ is taken over the predictor states S_t of the training transitions and so excludes the terminal state of the window. Here 500 kt is the multi-start initialiser rather than the lower bound, consistent with the frozen specification's constraint of K above the training maximum. The remaining estimation bounds are $\mathfrak{s} \in [0, \max_{\text{train}} S_t]$ with the further feasibility restriction $0 < \mathfrak{s} < 0.8K$ enforced in the objective, ϕ clipped to $[-0.95, 0.95]$, and the index exponent b fitted jointly with r and K without bounds. The profile of Section 3.2 restricts $\mathfrak{s}$ further, to $[0, \min_{\text{train}} S_t]$, for the reason given there. The reported fits attain the upper endpoint ($K = 5000$ kt) where the objective is flat out to the imposed bound; M1b's recovery-window $K = 105.8$ kt is a valid interior fit, not a bound violation. Where a reported fit sits at 500.0 kt this is the multi-start initialiser, which the optimiser has not left because the objective is flat there, and not an active constraint. M3's autoregressive coefficient is estimated in a second stage: r and K are fitted first by one-step least squares, and ϕ is then obtained as the lag-one regression coefficient of the resulting training residuals on their own lag, clipped to $[-0.95, 0.95]$; it is not jointly optimised with r and K. Every fitted-parameter value this article prints, with the window it belongs to and the bound it attains, is collected in Table 10; the per-origin rolling fits and the parameters the article does not print are archive items, marked there as such.

A survey-start variant replaces the surplus initial condition by $\hat{q}\, I_t$, where I_t is the autumn research-vessel abundance index and $\hat{q}$ is the training-window median of SSB/I. The index is an NCAM input, not an independent stock.

Prey-informed variants (Section 3.4) scale surplus production by a 1991 capelin regime or by the tabulated 3L spring acoustic index (DFO, 2024b). Pre-1991 acoustic values are not carried across 1991. Missing survey years are not interpolated; within the post-1991 regime the last eligible observation is carried forward, and training transitions with no eligible index value are dropped from the fit.

2.3 Evaluation design

What the design does and does not test should be stated before the scores. The predictand at every origin is a single retrospectively reconstructed assessment series, not the assessment vintage that would have been available at that origin; assessment vintages were not available for this evaluation. Earlier SSB values therefore embed information produced after the nominal forecast date, and the smoothing inherent in an assessment reconstruction is present in both the target and the persistence baseline. Catch is supplied along the forecast horizon from the catch file rather than forecast. The exercise is accordingly a retrospective prediction experiment and, with respect to catch and the prey index, a conditional hindcast; it is not a test of operational forecasting skill, and the comparison against persistence inherits the same status. Section 4 returns to what this does and does not license.

Fixed windows on Specification A: collapse, train 1983–1990, test 1991–1995; recovery, train 1995–2007, test 2008–2015. The year 1995 appears in the collapse test window and again as the first recovery training year. The two fixed-window exercises are separate fits scored separately and neither informs the other, so this is not leakage within an exercise; it is noted because the two windows are reported side by side. Rolling origin: minimum eight training years; horizons $h = 1$ and $h = 5$. Estimation is by one-step least squares on the training window, while the five-year score evaluates iterated trajectories of parameters fitted for one-step fit — a training/testing cost-function mismatch; trajectory-matched estimation over the multi-year horizon is an untested alternative, not scored here.

The primary score is RMSE of SSB (kt), taken over all scored origin–target pairs at the stated horizon; the $h = 5$ score is the root mean square of the five-year-ahead errors themselves, not an average along the intervening trajectory. Log-RMSE uses natural logarithms of states floored at 10^{-3} kt. The sign-hit rate compares $\mathrm{sign}(\Delta S)$ between forecast and observation over adjacent years within a window, giving one fewer comparison than there are scored years, and is reported as NA where a rule supplies no directional prediction rather than as a zero hit rate. Secondary scores are mean absolute error, RMSE on $\log S$, the threshold misclassification rate for $\mathbf{1}\{\hat{S} < \mathrm{LRP}\}$ — equivalently the Brier score for a deterministic binary forecast, which is what a hard 0/1 prediction makes it, so no probability forecast is implied — and the sign-hit rate of ΔS on fixed windows. On Specification A the Brier indicator rests on a degenerate event — the 884.6-kt LRP sits above the entire origin range, so the indicator is 1 at every origin and persistence attains 0.00 by construction. The score does separate models in this sample (persistence and the training mean at 0.00 against 0.80–1.00 for the structural modules, Table 3), but it separates them on an outcome that never varies, so it carries no early-warning information and is reported as a secondary diagnostic only (Section 3.3 gives the record). The sign-hit rate is conventionally 0.00 for persistence, whose forecast $\Delta S = 0$ has no sign; persistence is excluded from that ranking by declaration.

The retention rule is stated as explicit hypotheses.

Retention rule. *A module M on the ladder of Table 2 is retained only if all of the following hold on the rolling-origin primary RMSE score:*

- *(H1) M reduces primary RMSE relative to its declared comparator — the next-simpler rung for the nested steps (M1b against M1, M3 against M2); for the two rungs that are not strict nestings, M1 for M2 (the autonomous constant-catch map whose catch treatment M2 changes) and M3 for M4 (the module the delay acts on). M1b is reported as the alternative comparator for M2 and never decides retention. M1 is the first structural rung and has no simpler structural comparator, so H1 is vacuous for M1 and its retention turns on H2 and H3;*
- *(H2) M reduces primary RMSE relative to last-value persistence;*
- *(H3) each required reduction holds at both horizons $h = 1$ and $h = 5$ — the frozen specification's primary score is the rolling-origin RMSE pair — and exceeds a tie band of 5% of the comparator's score; improvements inside the band are ties and do not retain.*

Retention is decided separately on each specification. A module failing any of (H1)–(H3) is not retained. Two disclosures complete the rule. First, it is a point-rule ranking under a pre-set score, not a test of equal predictive ability; the uncertainty analysis of Section 3.5 attaches Diebold–Mariano and moving-block-bootstrap intervals to its margins and changes no verdict. Second, the tie band and the comparator declarations are completions recorded after the scores of Section 3 were computed: no recorded verdict depends on them — the smallest structural deficit against persistence (M1b's on Specification A at $h = 1$) is 17%, and no (H1) comparison in Tables 3–8 reverses under either comparator reading.

A module is not retained when the one-step least-squares implementation evaluated here fails at least one of (H1)–(H3) on the rolling-origin primary RMSE score on the series tested. Non-retention is a statement about these implementations on these series, not about every possible implementation of the underlying model class, and it is weaker than a statistical null result: it records that a module did not clear a stated bar, not that no difference exists.

The scoring is deterministic: the scripts are listed in the Data availability statement with pinned output checksums and an archived environment specification, so the reported arithmetic regenerates byte for byte within that environment. Cross-environment numerical identity is not claimed, and one M1b row is environment-sensitive at the ±17-kt level, as recorded there. Reproducibility of the arithmetic is distinct from the analytical result of Section 4, which is proved rather than computed.

On Specification B the collapse window is train 1954–1989, test 1990–1995, and the recovery-stall window is train 1995–2012, test 2013–2024. Catch is Table 1 landings (2024 persisted from 2023). No row is taken from NCAM 2016. The ladder's rolling origin on Specification B uses a minimum of twelve training years ($n = 59$ at $h = 1$, $n = 55$ at $h = 5$); the naive baselines retain the eight-year minimum ($n = 63$ and $n = 59$). The two origin sets therefore overlap but are not identical, so a baseline computed over the longer naive origin sets would mix four extra early origins into the comparison. Table 6 accordingly reports the origin-matched baselines, computed on the ladder's own origins: on the identical twelve-year origins persistence reads 84 kt ($n = 59$) at $h = 1$ and 300 kt ($n = 55$) at $h = 5$, and the training mean 458 kt and 522 kt, against M1's 120 kt and 432 kt. Persistence remains the lowest-RMSE forecast. The mixed-origin reading (88 kt, quoted in the Table 6 caption) exceeds the controlled reading of 84.4 kt by 3.6 kt — an origin-mix effect of no retention consequence. On the identical post-break origins of the Table 7 control ($n = 33$, $h = 1$) the recomputed baseline is 75 kt, matching the post-break value reported with the two-regime control of Section 3.4. On the index modules' origin sets (Table 8), persistence reads 97 kt ($n = 24$, $h = 1$) and 193 kt ($n = 20$, $h = 5$) on Specification A and 79 kt ($n = 36$) and 288 kt ($n = 32$) on Specification B; the index module loses to the origin-matched baseline in every cell, and the verdict — the module is not retained — is unchanged.

3. Results

3.1 Primary specification

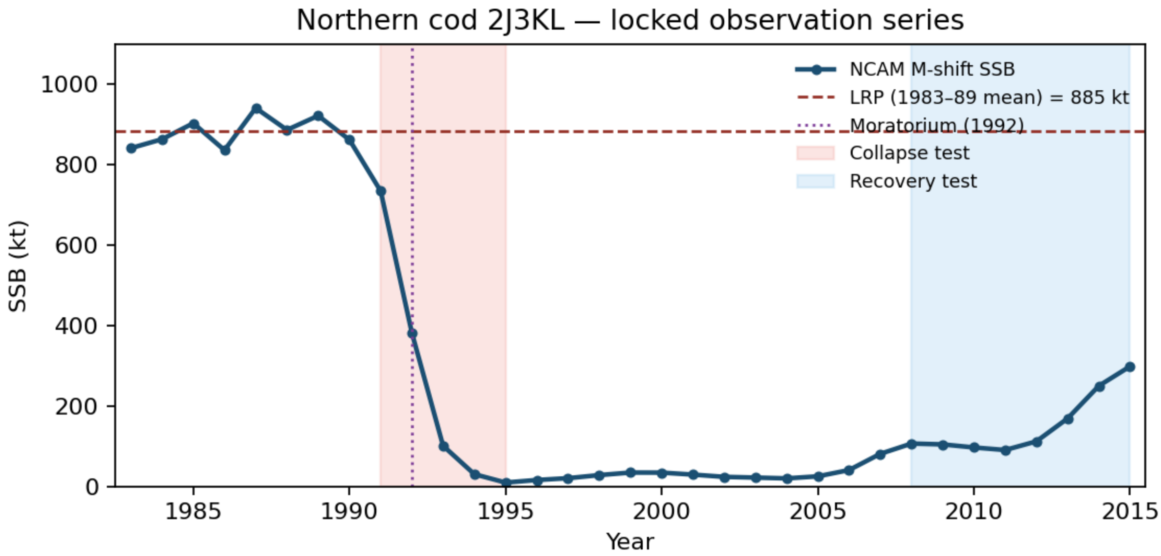

Figure 1: NCAM M-shift SSB (DFO, 2016, Table A2). Dashed line: 2016 LRP = 884.6 kt. 2015 SSB is 33.8% of that LRP, matching the advisory report statement of 34%.

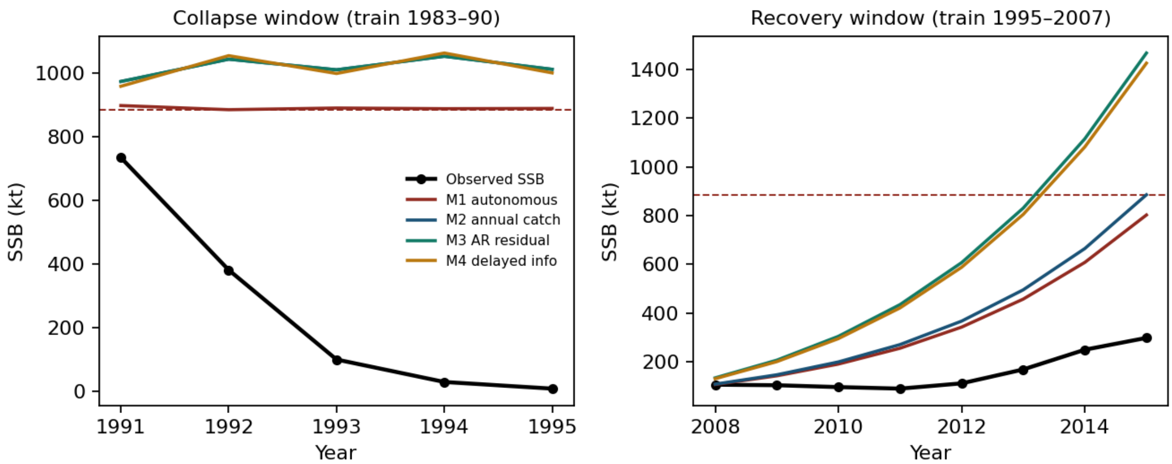

Figure 2: Multi-step forecasts from the end of each training window. Recovery is under-predicted when an AR residual fitted on a slow early-recovery train is persisted.

Table 3. Fixed-window scores (RMSE in kt), with both baselines reported on every window, computed on the same fixed origins set in Section 2.3. Table 3 is not to be read as a retention table — retention is decided on rolling-origin primary RMSE only.

Window	Model	RMSE	MAE	log-RMSE	Brier	Direction
Collapse	persist	670	610	2.71	0.00	0.00
	M1	694	638	2.73	1.00	0.50
	M1b	694	636	2.73	0.80	0.00
	M2	819	751	2.85	1.00	0.25

Window	Model	RMSE	MAE	log-RMSE	Brier	Direction
	M3	819	751	2.85	1.00	0.25
	M4	819	750	2.85	1.00	0.25
	mean	688	630	2.72	0.00	0.00
Recovery	persist	104	72	0.69	0.00	0.00
	M1 = M2	120	105	0.61	0.00	0.57
	M1b	**90**	55	0.52	0.00	0.57
	M3	220	200	0.92	0.00	0.57
	M4	214	195	0.91	0.00	0.57
	mean	144	124	1.60	0.00	0.00

On collapse, fitted r reaches 1.935, close to but not at the imposed upper bound of 2. The 1983–1990 window is a high, weakly trending stock. The fitted pre-collapse model does not reproduce the subsequent decline, and dropping C from 240 to 5 raises forecast SSB. The 1992 catch drop is an endogenous response to depletion rather than an exogenous policy lever; prescribing it in a constant-r stock-flow equation forces the mechanical rebound instead of reproducing the crash. Within this accounting, had the crash been driven by the catch regime one would expect supplying the catch path to help; M2 does not improve on M1. Supplying the prescribed catch path did not improve this implementation's collapse-window hindcast. That is not proof of one alternative: a misspecified and weakly identified module need not improve even when the variable it adds is genuinely relevant, so the result bounds what this ladder can attribute rather than identifying the mechanism.

On recovery, M1 and M2 coincide. The reason is structural:

On the recovery window (train 1995–2007, test 2008–2015) of Specification A, M1 and M2 produce identical forecasts: M1 plugs the training-mean catch into the autonomous map and M2 prescribes C_t year by year, but $C_t \equiv 5$ kt on both train and test windows, so the two prescriptions coincide and the least-squares (r, K) optimum is identical.

M1b reports a lower RMSE (90 kt against M1's 120 kt), but $\mathfrak{s} \to 0$ and K collapses to the training range: an unidentified Allee parameter, not a biological threshold. The estimate sits at the boundary for a reason visible in the objective.

Observation 3.2 (Boundary Allee estimate on the recovery window). Any $\mathfrak{s}$ strictly above the training minimum makes the depensation factor $a(S_t) = (S_t - \mathfrak{s})/(K - \mathfrak{s})$ negative for observations below $\mathfrak{s}$, so wherever the observed ΔS is positive at such a state the fit incurs a large squared residual. On the 1995–2007 recovery window seven of the twelve annual increments are positive — the five negative ones fall in a single consecutive run, 1999–2004 — and on balance this penalty pushes the estimate downward: the fitted $\mathfrak{s}$ approaches its lower bound ($\mathfrak{s} \to 0$; 2.1×10^{-23} under the coarse regime and 9.4×10^{-6} under annual landings) while r reaches its upper bound of 2 and K collapses to the training range.

This is an empirical finding about these fits, not a theorem, and we do not state it as one. Two qualifications matter. The pressure just described is a tendency and not a proof that the global optimum must lie below the training minimum: the objective trades errors across all observations, and r, K, $\mathfrak{s}$ and C can compensate for one another. The window is also not monotone — SSB declines in five consecutive years, from 34.59 kt in 1999 to 20.07 kt in 2004, before rising to 81.10 kt in 2007 — so no argument that assumes a monotone recovery applies to it. Nor is a boundary estimate by itself proof of non-identifiability, and a biologically meaningful threshold may lie outside the sampled range.

A profile of the least-squares objective shows how weakly the data constrain $\mathfrak{s}$. The grid for $\mathfrak{s}$ spans the estimation range $[0, \min_{\text{train}} S_t] = [0, 9.68]$ kt. That upper limit is the bound imposed in the code, chosen because the depensation factor turns negative on part of the window above it; it is not a claim that larger thresholds are mathematically inadmissible. Fixing $\mathfrak{s}$ on that grid and re-optimising r and K at each value gives a profile objective that rises only from 126.50 to 135.04 kt^2 under the coarse regime, and from 122.90 to 126.50 kt^2 under annual landings.

Expressed as $n_{\mathrm{tr}} \log\{\mathrm{SSE}(\mathfrak{s})/\mathrm{SSE}(\hat{\mathfrak{s}})\}$ with $n_{\mathrm{tr}} = 12$, the largest deviation across the grid is 0.78 under the coarse regime and 0.35 under annual landings. We report these as descriptive measures of curvature rather than as calibrated likelihood-ratio tests: the least-squares fit carries no error model, the estimate $\hat{\mathfrak{s}}$ lies on the boundary of its range, r rests at its own upper bound, the window supplies twelve transitions, and the states are assessment reconstructions rather than direct observations. Any one of these would compromise a χ^2_1 reference. The substantive reading does not depend on the reference distribution: the objective is nearly flat across the whole range, so the data do not locate $\mathfrak{s}$ within it.

The mechanism is parameter compensation. In $a(S_t) = (S_t - \mathfrak{s})/(K - \mathfrak{s})$, raising $\mathfrak{s}$ reduces the numerator while lowering K raises $1/(K - \mathfrak{s})$, and along the profile K slides from 106.0 to 82.7 kt (coarse) or 129.8 to 95.6 kt (annual landings) while r remains at its upper bound of 2 throughout. The product $rS_t(1 - S_t/K)\, a(S_t)$ is nearly invariant along this path, so the data cannot separate $\mathfrak{s}$ from K. The training observations therefore do not constrain behaviour near a putative threshold, and the boundary estimate is not evidence about a biological Allee threshold in either direction.

M3's $\phi = 0.95$ persists a negative residual and increases error.

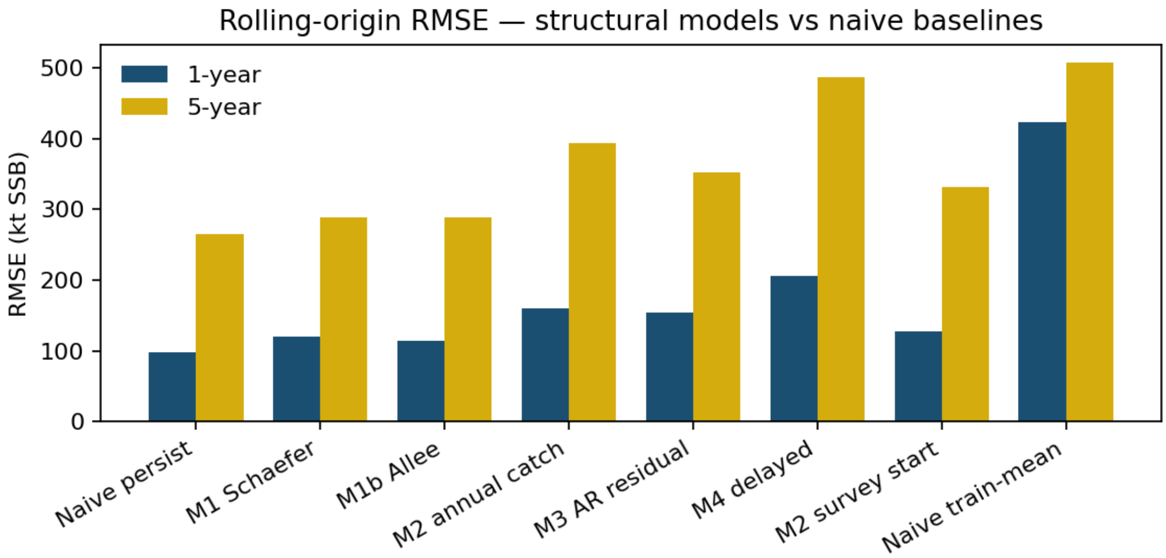

Figure 3: Rolling-origin RMSE. Persistence is the best one-year and five-year point forecast.

Table 4. Rolling-origin summary, Specification A, coarse catch regime.

Model	$h = 1$ RMSE	$h = 1$ MAE	$h = 1$ log-RMSE	$h = 5$ RMSE
persist	**98**	**48**	**0.52**	**265**
M1	121	80	8.02	289
M1b	115	80	8.70	289
M2	144	61	0.59	398
M3	135	53	3.39	366
M4	196	82	0.76	488
mean	424	375	2.35	507

Log-RMSE for M1 and M1b is large because some origins produce trajectories that hit the numerical floor. M2 keeps the state off zero and stabilizes the log score, but loses on raw RMSE. M4, the only model started from a year-old state, is the worst structural model on raw RMSE; its decomposition against a stale-start persistence control is reported in Section 4.

No structural model — M1 through M4 — is retained on Specification A. Persistence is the lowest-RMSE forecast.

3.2 Annual landings and survey start

Year-by-year landings for 2015 are 4.436 kt, matching DFO reported landings. Pre-collapse catches are 172–269 kt, not a flat 240. The 1992 drop is to 41 kt, then 11 kt, then 0.4–1.9 kt.

Table 5. Rolling-origin RMSE (kt) with annual landings.

Model	$h = 1$	$h = 5$
persist	**98**	**265**
M1	121	289
M1b	115	289
M2	160	394
M3	154	352
M4	206	486
M2, survey start	128	331

Annual landings make M2 worse than the coarse regime on one-year RMSE (160 versus 144 kt). Collapse-window RMSE under annual landings remains about 821 kt for M2, M3 and M4 (819 kt under the coarse regime; M1 and M1b are at 694 kt on both treatments, Table 3). On the annual-catch recovery window (train 1995–2007, test 2008–2015) the ladder scores M1 264, M1b 78, M2 303, M3 609, and M4 586 kt, against a frozen-persistence error of 104 kt (forecast held at the 2007 level of 81 kt). Annual landings therefore do not repair the recovery-window fit, and the residual-persistence models deteriorate severely (609 and 586 kt against 220 and 214 kt under the coarse regime). M1b's 78 kt carries the same unidentified-Allee caveat as its regime-window fit.

Apparent net production $S_{t+1} - S_t + C_t$ is strongly negative in 1991–93 even after subtracting reconstructed catch. This quantity is a diagnostic construct, not an exact stock balance, and we do not treat it as one: S here is spawning-stock biomass while C_t is total landings, and changes in SSB also reflect maturation into the spawning stock, changing maturity and weight-at-age schedules, and the age composition of catch and of natural deaths. Within the surplus-production accounting the ladder itself assumes — in which S is the modelled biomass state and C_t is subtracted from it — matching the observed ΔS with no production at all would require removals of order $-\Delta S$, and the observed decline is far larger than C_t. On that reading the unexplained term dwarfs the difference between the two catch treatments. Regular et al.'s $M \approx 2.5$ peak is an age-structured assessment's estimate informed by additional data and assumptions; it is not the same object as this scalar residual, though the two point in the same direction. Within the model class tested, substituting annual recorded landings for the coarse regime does not produce the crash in a constant-r surplus model.

One numerical disclosure travels with the window: the annual-landings M1 (264 kt) and M1b (78 kt) differ from the coarse-regime recovery rows (M1 = 120, M1b = 90) although both treatments sit on the same SSB column. The reconciliation is that M1's constant catch is not estimated on that column. It is the training mean of the catch file: 5.0 kt for the coarse regime, and 3.19 kt (the 1995–2007 mean of the annual landings) for the annual treatment. The two least-squares objectives are nearly flat at their minima (MSE 128.35 versus 127.84 kt^2), so the (r, K) minimizer slides along the valley as the constant changes: $r = 0.458$ with K resting at the multi-start initialiser of 500.0 kt — not a bound, the lower bound on this window being 50.8 kt — against $r = 0.370$ with K at its upper bound, 5000.0 kt. The optimiser leaves K at its starting value in this fit; the profile of Section 3.2, not the optimiser's behaviour, is what measures the flatness. The same geometry explains the M1b pair (r pinned at its upper bound in both treatments; $K = 105.8$ against 129.8 kt): the dependence of the M1 fit on the catch file is the flat-objective identification of a two-parameter autonomous fit on a 13-year window, not an inconsistency between fitting objects. The same flat valley caps what the ranking can resolve. Refitting the recovery-window Schaefer map with K fixed anywhere on $[60, 5000]$ kt and r re-optimised moves the one-step training objective only from mean squared error 127.4 to 149.9 kt^2 (training RMSE 11.29–12.24 kt, a spread under 0.95 kt), while r compensates from 0.435 to 0.773. On this window the (r, K) pair is not identified, and the ordering of valley variants is not a robust ranking. One mechanism unifies both faces of the catch-dependence: a constant catch shifts the flat valley's minimiser without moving the objective, so the rolling M1 and M1b scores (refit at every origin) are insensitive to the catch treatment — both are identical to the kilotonne across the two treatments (121 and 115 kt at $h = 1$) — while the single fixed-window fit is not. The property is specific to the two constant-catch modules: M2, M3 and M4 take C_t into the state update at every step, and their one-year rolling scores do move with the treatment (144 to 160, 135 to 154, and 196 to 206 kt respectively).

Starting from $\hat{q}I_t$ instead of SSB gives one-year RMSE 128 kt (still worse than persist 98); one-year log-RMSE 0.49 versus persist 0.52; five-year RMSE 331 versus persist 265. On the primary score the variant is not retained. The log-score difference is recorded and not used for selection. The survey index contributes to NCAM and therefore does not provide an independent validation target.

3.3 Alternative assessment specification

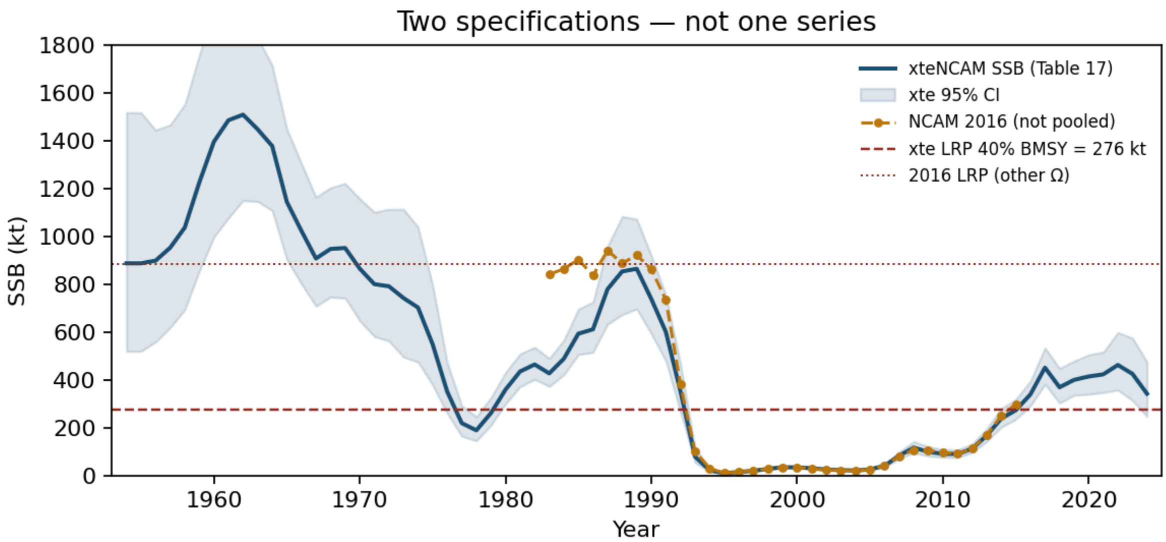

Figure 4: The two specifications. Overlap 1983–2015 RMSE = 126 kt (2015: NCAM 299 kt, xteNCAM 273 kt). Different safe sets. Not pooled.

Checkpoints from Regular et al. (2025) Table 17: 2005 SSB = 26 kt (95% interval 22–31), 2017 = 451 kt (381–534), 2021 = 423 kt (NCAM and xteNCAM said to agree), 2024 = 342 kt (246–475), 2024/LRP = 1.24. The 2005 values (26 kt versus NCAM 25.18 kt) can agree on a low year without the two series sharing a reference point; agreement on a low year does not warrant splicing the two columns. The same late-period biomass is 34% of the old LRP and above the new one after 2016.

Table 6. Rolling-origin RMSE on Specification B (kt). The persistence and training-mean rows are the origin-matched baselines, computed on the ladder's own origin sets ($n = 59$ at $h = 1$, $n = 55$ at $h = 5$); the mixed-origin baselines over the longer naive origin sets ($n = 63$ and $n = 59$) are 88 and 318 kt for persistence and 449 and 506 kt for the training mean.

Model	$h=1$	$h=5$
persist	**84**	**300**
M1	120	432
M1b	152	446
M2	166	1059
M3	127	930
M4	206	1031
mean	458	522

Collapse window (train 1954–89, test 1990–95): M1 RMSE 817 kt; M2 1898 kt. Official landings worsen the crash forecast. No module is retained. Persistence remains the lowest-RMSE forecast. This is a second non-retention outcome under the same rule on a different specification, and it is weaker than a statistical null result. The longer series and the revised LRP do not justify retaining the additional modules.

Recovery-stall window (train 1995–2012, test 2013–2024; $n = 12$): M1 254 kt, M1b 178 kt, M2 269 kt, M3 268 kt, M4 269 kt. The 2013–2024 test path rises from 167 kt to a 2022 peak of 462 kt before easing to 342 kt in 2024, all above the 112 kt training-end value, so a persistence forecast frozen at that value errs by 262 kt — more than M1 and M1b on this single origin. This fixed-origin comparison does not enter the retention rule, which is decided on rolling-origin primary RMSE (Table 6, where persistence is not beaten at either horizon). M1b's window fit carries the identification fragility familiar from its other window fits: a small fitted Allee parameter ($\mathfrak{s} = 5.3 \times 10^{-3}$) with a carrying capacity ($K = 500$ kt) extrapolated far beyond the training range (maximum 117 kt).

On the rolling Brier score for $\mathbf{1}\{\hat{S} < \text{LRP}\}$ on Specification B (the secondary score, LRP = 276 kt): persistence 0.06 at h=1 and 0.27 at h=5; M1 0.05 and 0.45; M1b 0.15 and 0.47; M2 0.07 and 0.36; M3 0.02 and 0.31; M4 0.07 and 0.38. M1 and M3 improve the one-year Brier score over persistence (0.05 and 0.02 versus 0.06); no model improves it at h=5, and no model improves the primary RMSE score at either horizon (Table 6), so the retention verdict is unchanged. For reference, the same secondary score on Specification A is 0.00 at both horizons for persistence. The persistence indicator forecasts $\mathbf{1}\{S_t < 884.6\}$ from the origin state, and its score is zero because every origin state is already below the 884.6-kt LRP. In particular the first origin, S_{1990}, lies below the 1983–89 mean bound, so the collapse train starts on the downslope of its own reference period. The target years 1991–2015 are below the LRP as well, so indicator and outcome agree at every origin. (The bare fact that targets are below the LRP would not by itself make the persistence indicator correct; the origin states being below it does.) Persistence supplies no directional prediction, so its sign-hit rate is reported as NA rather than as a zero hit rate. The structural scores are M1 0.04/0.05, M1b 0.00/0.05, M2 0.08/0.14, M3 0.08/0.10, M4 0.08/0.14 — unbeatable there, and reported only to keep the two specifications' records separate and complete.

Regular et al. assign the 1992–94 disappearance primarily to M (peak ≈ 2.5), informed by tagging and a capelin/cod predictor, and note that some of that M could still be unreported F. The relation between that attribution and the scalar residual of this ladder is stated in Section 3.2.

3.4 Prey-informed productivity

Murphy et al. (2025) report a 1991 acoustic collapse (1985–90 median 3704 kt versus 1991–2022 median 174 kt). A two-regime r with break 1991 uses the high regime only for forecasts issued before 1991. Capelin enters as a candidate disturbance class, not as a fitted parameter.

Table 7. Rolling RMSE, two-regime r.

Specification	Model	$h = 1$	$h = 5$
A	persist	**98**	**265**
A	M_cap	154	334
B	persist	**88**	**318**
B	M_cap	147	894

On post-break origins only (Specification B, $h = 1$; $n = 33$): M_cap 107 versus persistence 75 on the identical origin set (the 88 of Table 7 is the all-origins value). Not retained. On Specification A post-break origins ($n = 24$) the corresponding M_cap value is 149 kt.

The year-by-year 3L spring acoustic biomass is tabulated in the Northwest Atlantic Fisheries Centre's Zenodo record (2025), 10.5281/zenodo.17515115, with 2023 = 331.3 kt from Murphy et al. (2025). Surplus is scaled by $(I_{\text{known}}/I_{\text{ref}})^b$, where $I_{\text{known}}(t)$ is the last observation at or before t, subject to the regime rule of Section 2.2: the search does not reach back past 1991, so if no observation exists in $[1991, t]$ the module is not evaluated at that origin rather than falling back on a pre-1991 value. Along a multi-step trajectory t is the forecast year being stepped to, not the origin, so a five-year path may use index values published after its issue date; this is a conditional hindcast with respect to the index and is not an operational forecast. I_{ref} is the training-window median of the observed index, and b is a third free parameter fitted jointly with r and K by one-step least squares on the training window. The module additionally faces degrees-of-freedom starvation: three parameters (r, K, b) are fitted on as few as eight training years at the shortest origins, which invites in-sample overfitting and penalizes the one-year out-of-sample comparison. Survey years are unobserved in part of the record, so the index-forecast origins number $n = 24$ ($h = 1$) and $n = 20$ ($h = 5$) on Specification A and $n = 36$ and $n = 32$ on Specification B, against the SSB-based rolling-origin counts ($n = 25$ at $h = 1$ and $n = 21$ at $h = 5$ on Specification A; Section 4).

Table 8. Rolling RMSE, observed acoustic index. Bold marks the best mixed-origin baseline in each column. At $h = 5$ on Specification A the module (262 kt) reads below the mixed-origin baseline (265 kt); that apparent 3-kt advantage is an origin-mix artefact and is not marked, because the origin-matched baseline on the module's own origins is 193 kt. The comparison that matters is the matched one.

Specification	Model	$h = 1$	$h = 5$
A	persist	**98**	265
A	persist (origin-matched)	97	193
A	M_cap_index	150	262
B	persist	**88**	**318**
B	persist (origin-matched)	79	288
B	M_cap_index	132	492

On the origin-matched reading the five-year near-tie on Specification A dissolves — the baseline on the module's own origins reads 193 kt against the module's 262 kt — and one-year RMSE is worse under both readings (150 versus 98 kt on the SSB origins and 97 kt on the module's); on Specification B the module's 132/492 kt stand against origin-matched baselines of 79/288 kt. The module is not retained under either baseline reading.

3.5 Uncertainty on the retention margins

The uncertainty analysis attaches Diebold–Mariano tests (Diebold and Mariano, 1995; unweighted HAC truncation at lag $h - 1$) and moving-block bootstrap intervals (Künsch, 1989; block length $\max(h, 3)$, 20,000 replications, seeded) to the retention margins of the retention rule. It is computed from the archived per-origin forecast files: on Specification B the xteNCAM rolling file; on Specification A the archived per-origin file is the annual-landings pass of Section

3.2 (the coarse-regime pass's per-origin rows are not archived — its summary is — and the verdicts coincide under both treatments). The persistence baseline is not archived per-origin; it is recomputed on the identical origin sets from the registered series, and the recomputation reproduces the recorded origin-matched baselines (98 and 265 kt on Specification A; 84 and 300 kt on the matched Specification B origins) — the script fails loudly otherwise. The layer attaches uncertainty to margins the point rule has already ranked; it changes no frozen verdict, score, or table value.

The two columns of Table 9 summarise the same comparison through different approximations. The DM statistic z tests the mean of the per-origin squared-loss differential $d_i = L_{A,i} - L_{B,i}$ against zero, scaled by a HAC standard error. The interval and p come from the moving-block bootstrap of the difference of RMSEs, $\sqrt{\overline{L_A}} - \sqrt{\overline{L_B}}$. The two effect measures are monotonically related, since $\sqrt{\overline{L_A}} - \sqrt{\overline{L_B}} = (\overline{L_A} - \overline{L_B})/(\sqrt{\overline{L_A}} + \sqrt{\overline{L_B}})$, so they share a sign and a zero null; the square root rescales the gap but does not by itself make one summary more stable than the other. What differs is the approximation used to gauge sampling variability — a HAC-scaled asymptotic normal reference against a resampling distribution over blocks — and in samples this small and this dependent the two need not agree. Four of the twenty-eight rows have an interval excluding zero while $|z| < 1.96$ (the rows are named in Section SI-3). We therefore report the two summaries side by side and do not treat the bootstrap interval as a second calibrated test. The interval and the p are mutually consistent in every row. We also note that p is obtained by inverting the percentile interval rather than from a null-centred resampling scheme, so it is an interval-based summary and not a calibrated test of a sharp null; where the two columns disagree we report both and rest no verdict on either.

Because the origins overlap at $h = 5$, the estimation window expands, the predictand is a smoothed reconstruction and the sample straddles a structural break, the automatic $h - 1$ HAC truncation and the $\max(h, 3)$ block length should not be taken as neutral choices. The M1-against-persistence margin was therefore recomputed over a grid of both (Section SI-3). The DM statistic crosses the conventional threshold within the range of defensible bandwidths, upward on Specification A and downward on Specification B, so nothing rests on it; the bootstrap is markedly more stable across block lengths 3 to 9. The qualitative reading — Specification A within noise, Specification B separating — is therefore a bootstrap result rather than a DM result, established for that comparison and not asserted for every module. Read "within noise" as "this analysis does not resolve the difference", not as evidence that the models are equivalent.

Table 9. Uncertainty on the retention margins. Specification A rows are computed on the annual-landings pass of Section 3.2, not on the primary coarse-regime pass, whose per-origin files were not archived; the Specification A RMSE values below therefore reproduce Table 5 rather than the frozen Table 4. Specification B rows are the primary pass. (Diebold–Mariano with HAC lag $h - 1$; Künsch moving-block bootstrap, block length $\max(h, 3)$, 20,000 replications, seeded). Gaps are module minus comparator, in kt; positive gaps mean the module is worse.

Spec	h	Module	Comparator	n	RMSE module	RMSE comp.	Gap (kt)	DM z	95% CI (kt)	p
A	1	M1	persist	25	120.5	98.0	+22.5	1.15	[−13.3, +50.5]	0.192
A	1	M1b	persist	25	114.8	98.0	+16.7	0.98	[−19.3, +59.2]	0.376
A	1	M2	persist	25	160.4	98.0	+62.4	1.30	[−11.9, +107.0]	0.445
A	1	M3	persist	25	153.6	98.0	+55.6	1.02	[−18.0, +91.6]	0.927
A	1	M4	persist	25	206.3	98.0	+108.2	1.51	[−8.2, +192.7]	0.338

Spec	h	Module	Comparator	n	RMSE module	RMSE comp.	Gap (kt)	DM z	95% CI (kt)	p
A	1	M2	M1	25	160.4	120.5	+39.9	1.13	[−59.1, +88.5]	0.687
A	1	M4	M3	25	206.3	153.6	+52.7	0.99	[+4.7, +144.7]	<0.001
A	5	M1	persist	21	288.7	264.7	+24.0	1.84	[−43.1, +56.4]	0.279
A	5	M1b	persist	21	288.6	264.7	+23.9	1.84	[−43.1, +56.4]	0.284
A	5	M2	persist	21	393.8	264.7	+129.1	1.10	[−25.0, +215.9]	0.576
A	5	M3	persist	21	351.5	264.7	+86.8	1.10	[−7.3, +131.9]	0.266
A	5	M4	persist	21	486.4	264.7	+221.7	1.14	[−27.0, +412.8]	0.759
A	5	M2	M1	21	393.8	288.7	+105.1	0.97	[−78.8, +230.1]	0.807
A	5	M4	M3	21	486.4	351.5	+134.9	1.15	[−28.4, +292.4]	0.902
B	1	M1	persist	59	119.5	84.4	+35.0	2.81	[+2.7, +70.8]	0.032
B	1	M1b	persist	59	151.6	84.4	+67.2	3.60	[+18.3, +117.7]	0.009
B	1	M2	persist	59	166.0	84.4	+81.6	3.22	[+36.4, +122.8]	<0.001
B	1	M3	persist	59	126.7	84.4	+42.3	1.85	[+1.0, +92.5]	0.042
B	1	M4	persist	59	205.7	84.4	+121.3	3.20	[+53.7, +193.0]	<0.001
B	1	M2	M1	59	166.0	119.5	+46.6	1.78	[−20.5, +104.5]	0.170
B	1	M4	M3	59	205.7	126.7	+79.0	3.07	[+35.6, +116.4]	<0.001
B	5	M1	persist	55	431.9	300.0	+131.9	2.06	[+33.2, +218.6]	0.008
B	5	M1b	persist	55	445.5	300.0	+145.5	1.80	[+18.2, +250.9]	0.023
B	5	M2	persist	55	1058.9	300.0	+758.9	2.15	[+363.6, +1038.5]	<0.001
B	5	M3	persist	55	930.1	300.0	+630.2	2.39	[+352.8, +845.6]	<0.001
B	5	M4	persist	55	1030.7	300.0	+730.8	2.35	[+407.3, +978.3]	<0.001
B	5	M2	M1	55	1058.9	431.9	+627.0	1.97	[+195.7, +929.7]	0.004
B	5	M4	M3	55	1030.7	930.1	+100.6	1.88	[+20.2, +177.4]	0.007

Readings. On Specification A no non-retention margin against persistence separates from zero. At $h = 1$ the deficits against persistence (16.7–108.2 kt on $n = 25$) carry DM z from 0.98 to 1.51 with bootstrap p from 0.19 to 0.93; at $h = 5$ (23.9–221.7 kt on $n = 21$) z runs from 1.10 to 1.84 with p from 0.27 to 0.76. The heavy-tailed collapse-window losses dominate the bootstrap variance at this sample size, so the Specification A outcome is a point-rule ranking whose margins lie within noise: the scores suffice to order the models and do not resolve small skill differences. On Specification B the non-retention separates: every module's deficit against persistence has a bootstrap interval excluding zero at $h = 1$ (p from 0.042 to below 0.001 on $n = 59$) and at $h = 5$ (p at most 0.023 on $n = 55$). The declared (H1) comparators behave asymmetrically: M2's deficit against M1 is within noise at $h = 1$ on both specifications and at $h = 5$ on Specification A, separating only on Specification B at $h = 5$ ($p = 0.004$), while M4's delay cost against M3 separates at $h = 1$ on both specifications (p below 0.001) and at $h = 5$ on Specification B ($p = 0.007$) but not on Specification A ($p = 0.90$). The alternative comparator rows are held in the archive rather than printed in Table 9; on them the reading (M2 against M1b) is within noise at $h = 1$ on both specifications and at $h = 5$ on Specification A, separating on Specification B at $h = 5$ ($p = 0.005$): the comparator declaration of the retention rule is immaterial to every recorded verdict, as the protocol disclosure records. The layer is produced by a registered script, deterministic under seed 0, with its output archived alongside it; Section SI-3 names both.

3.6 Fitted parameters

This section collects in one place every fitted-parameter value the article prints, the window or treatment it belongs to, and the bound it attains. The table collects values reported in the preceding sections; nothing is recomputed. Quantities not reported individually (the per-origin rolling fits of every module, the index module's per-window exponent b, and the delay module's structural setting) are held in the archive rather than invented. The collection exists so the identification record — bounds attained, interior fits, the flat valley — can be read at a glance.

Table 10. Fitted parameters, with the section in which each is reported.

Module	Window / treatment	Fitted values	Bound / identification status
M1	Collapse, Specification A (train 1983–1990) (§1, §4)	$r = 1.935$, $K = 1032.7$ kt, constant catch 240 kt; repeller 144 kt, attractor 889 kt, monotone below 783 kt, $F'(S^*) \approx -0.39$	§3.1 records the collapse-window fitted r as constrained near, but not attaining, the upper bound of 2
M1	Recovery, coarse catch regime (§3.2)	$r = 0.458$, constant catch 5.0 kt (the training mean); training MSE 128.35 kt^2	K at 500.0 kt, the multi-start initialiser, not a bound (the lower bound on this window is $\max_{\text{train}} S + 10 = 50.8$ kt)
M1	Recovery, annual landings (§3.2)	$r = 0.370$, constant catch 3.19 kt (the 1995–2007 landings mean); training MSE 127.84 kt^2	K at its upper bound, 5000.0 kt
M1b	Recovery, coarse catch regime (§2.2, §3.2)	$\mathfrak{s} = 2.1 \times 10^{-23}$, $r = 2.000$, $K = 105.8$ kt	$\mathfrak{s}$ at its lower bound; r pinned at its upper bound in both treatments; K a valid interior fit
M1b	Recovery, annual landings (§3.2)	$\mathfrak{s} = 9.4 \times 10^{-6}$, $r = 2.000$, $K = 129.8$ kt	$\mathfrak{s}$ at the lower boundary to numerical tolerance
M1b	Recovery-stall, Specification B (train 1995–2012) (§3.3)	$\mathfrak{s} = 5.3 \times 10^{-3}$, $K = 500$ kt, training-window maximum 117 kt	identification fragility: $\mathfrak{s}$ small, K extrapolated far beyond the training range

Module	Window / treatment	Fitted values	Bound / identification status
M3	Rolling, Specification A (§3.1)	$\phi = 0.95$ (AR(1) residual coefficient)	printed value; the per-window rolling ϕ values are archive items
`M_cap_index`	Index module (§3.4)	r, K, b fitted jointly by one-step least squares	fitted values not printed (archive); degrees-of-freedom note at $n = 8$ training years
M1	Flat-valley sweep, recovery window (§3.2)	K fixed anywhere on $[60, 5000]$ kt moves the one-step training objective only from MSE 127.4 to 149.9 kt^2 (training RMSE 11.29–12.24 kt) while r compensates over $[0.435, 0.773]$	the (r, K) pair is not identified on this window; the ordering of valley variants is not a robust ranking
M4	Rolling, all specifications	r, K, ϕ refitted by the same procedure as M3 at every origin; the module adds no parameter of its own, the one-year delay being structural	archive: the refitted per-origin values coincide with M3's and are not printed separately
All modules	Per-origin rolling fits	archive items (not printed); M1b's Specification B rolling Allee optimum is environment-sensitive at the ± 17 kt level (§Data availability)	—

Bounds: $r \in (0.001, 2]$, and K optimised on $[\max_{\text{train}} S + 10, 5000]$ kt, where the maximum is taken over the states the one-step regression conditions on. On the 1995–2007 recovery window those are the years 1995–2006, with maximum 40.8 kt, giving a lower bound of 50.8 kt. Optimisation starts from $K = 500$ kt. The recovery-window M1 fit reports $K = 500.0$ kt: the optimiser does not move from its starting value, which lies 449 kt above the bound, because the objective is flat there. This is the same flat-valley geometry quantified in Section 3.2, and a stationary starting value is a sharper symptom of it than an active bound would be.

4. Discussion

The observed path falls far below the fitted lower equilibrium and then recovers. No exact fixed autonomous noise-free scalar trajectory can reproduce that. The obstruction is structural, and the scored collapse failure is its finite-sample face.

Proposition 4.1 (No recovery from below the lower equilibrium, for autonomous noise-free scalar surplus maps). *Let $S_{t+1} = F(S_t)$ be an autonomous one-dimensional surplus-production map of Definition 2.1 evaluated with $\varepsilon_t \equiv 0$ and a constant catch $C_t \equiv C$, with at most two positive equilibria — a lower repelling point S_- and an upper attracting point — and a one-step map monotone below its maximum; the single-equilibrium monotone regime is the special case, and the fitted collapse-window map ($r = 1.935$, $K = 1032.7$ kt, constant catch 240 kt) has exactly this form (repeller 144 kt, attractor 889 kt, monotone below 783 kt). If the observed path (S_t) falls strictly below S_- at some time and is strictly larger at some later time, then no trajectory of F reproduces (S_t) exactly. The Specification A collapse path satisfies the* *hypothesis: NCAM SSB falls to* 9.7 *kt by 1995, far below the fitted repeller of* 144 *kt, and rises thereafter. M1's collapse failure on Specification A is the finite-sample face of this obstruction.*

The argument is immediate: below the lower equilibrium $F(S) < S$, and $[0, S_-)$ is forward-invariant because F is monotone there and $F(S_-) = S_-$, the interval being closed at the left because the implementation clips the state at a non-negative floor. Every trajectory entering $[0, S_-)$ therefore decreases toward the floor and never returns above S_-, so a path that goes below S_- and is later higher is not a trajectory of F.

Two limits on the scope of this proposition should be stated explicitly, because both are easy to overstate. First, it does not say that autonomous maps cannot produce non-monotone paths: the fitted map has $F'(S^\star) \approx -0.39$ at its attractor, so approach to the attractor is by damped oscillation, and declines followed by recoveries are the generic local behaviour there (from 950 kt the fitted map gives $950 \to 857 \to 899$ kt). What is excluded is recovery after a fall below the lower repeller, which is the feature the observed collapse path actually has. Second, the result is a statement about autonomous noise-free scalar maps only. It does not apply to the stochastic map of Definition 2.1 with $\varepsilon_t \neq 0$, nor to M3 and M4, which carry a residual state and a stale initial state respectively and are therefore not autonomous scalar maps. The proposition applies directly to M1 under the deterministic forecast convention used here, and to M1b whenever its own fitted map has the stated equilibrium structure. It does not formally cover M2, whose prescribed year-by-year catch makes the update a time-indexed family F_t rather than a single autonomous map; for M2 the argument is motivating rather than binding, and the collapse-window result of Section 3.3 is the relevant evidence. It is not an impossibility theorem for surplus-production modelling in general.

The fixed-window sign-hit rates of ΔS (0.00–0.50 on collapse; Table 3) record the directional character of that failure: the structural models miss not only the magnitude of the crash but the sign of the trajectory exactly when the official catch series drops.

Under both the coarse catch regime and official landings, lowering C in 1992 cannot generate the crash. Collapse on these specifications is a productivity, unallocated-mortality, or observation event, not a surplus-production response to the catch drop. That is compatible with DFO's caution that NCAM M can absorb unreported deaths: the crash remains such an event under the best available public catch reconstruction, which tightens that limitation rather than relaxing it.

Unidentified extra structure does not close the gap to persistence, and in two of the four modules it widens it. The direction is not uniform and should not be stated as though it were: relative to its own declared comparator M3 reads below M2 in all six rolling comparisons (by 6 to 129 kt), and M1b reads below M1 at $h = 1$ on Specification A and on the fixed recovery window. What fails is H2, not always H1. Where the added structure does hurt, the mechanism is visible: autoregressive residuals fitted on short, regime-changing windows persist the wrong sign, an Allee parameter goes to the boundary, and a one-year delay removes information. These modules stay descriptive unless the conditions under which a module's contribution could be certified rather than merely scored hold.

The 2016 LRP is the 1980s mean SSB, so over its own reference years the mean deviation of SSB from that bound is zero by construction, though individual annual deviations are not. Defining the threshold from the same years that a leading-indicator claim would be tested on does not establish prospective early-warning performance for 1983–1990.

The moratorium is a change in implementable catch and is already in C_t. A separate delay switch does not add a degree of freedom beyond stock-flow. Whether the 1992 management architecture admits any feasible catch path that keeps the stock above its reference threshold is a separate question from one-year forecast error, treated for this stock in Abaee (2026c), and is not addressed here.

One-dimensional surplus production is not NCAM: age structure, migration, and survey catchability are omitted. The test is whether this ladder is retained under the stated score, not whether the assessment is a good filter. M4 is a stale-start experiment rather than a fully delayed information set: its trajectory begins at S_{t-1}, but its parameters are M3's, fitted on training transitions that end at S_t, so the later state still reaches the forecast through the fitted coefficients and the carried residual. A genuinely delayed variant would truncate training as well, and is not scored here. Its one-year gap against persistence therefore mixes a stale initial state with the model's own cost.

The separating control — persistence issued from the last available assessment, S_{t-1} — decomposes the gap. On Specification A under the coarse catch regime (the primary pass) the control reads 184.4 kt at $h = 1$ and 329.8 kt at $h = 5$, against M4's 195.6 kt and 488.3 kt. The information delay, the year-old start under the same persistence rule, therefore accounts for 86.4 of the 97.5-kt one-year gap and the surplus model's own cost for the remaining 11.1 kt; at $h = 5$ the split is 65.1 against 158.4 kt. Table 9's Specification A rows are the annual-landings pass, where M4 reads 206.3 kt, so the two sets of M4 figures are not directly comparable.

On Specification B every term is computed on M4's own origin set, so that the control, the baseline and the module share origins ($n = 59$ at $h = 1$, $n = 55$ at $h = 5$). Matched persistence reads 84.4 and 300.0 kt and the stale-start control 158.0 and 337.4 kt, against M4's 205.7 and 1030.7 kt. The delay accounts for 73.6 of the 121.3-kt one-year gap and the model's own cost for the remaining 47.7 kt; at $h = 5$ the model's own cost dominates (693.3 of 730.8 kt). These are descriptive comparisons between two forecast rules, not a causal decomposition of the value of assessment timeliness: the delayed control still reads off the same smoothed reconstruction, and access to a later reconstruction is not the same thing as access to a timely real-world assessment. With that reading, M4's error is mostly attributable to the information delay at the one-year horizon on both specifications, and mostly the surplus model's own structure at the five-year horizon on Specification B: at $h = 1$ the surplus model's own penalty is only about 11 kt — the delay, not the structure, separates M4 from persistence. The same arithmetic carries a policy reading: the one-year information delay costs 86.4 kt at $h = 1$ on Specification A, more than the structural cost of any delay-free module (M1 23, M1b 17, M2 46, M3 37 kt over timely persistence) and more than M4's own structure-given-delay cost of 11.1 kt. On these series the value of a timely assessment exceeds the value of any structure tested.

The predictand is an assessment smoother (NCAM/xteNCAM SSB), not a raw observation: forecasting it is forecasting a filtered state, and persistence inherits the smoother's autocorrelation — the comparison is valid for "does this ladder track this assessment," not for a vintage or early-warning reading, which would require assessment vintages at each origin, and none is available. At the five-year horizon the persistence baseline additionally carries a demographic reading that the evaluation does not assert: holding SSB constant over roughly a full cod generation implicitly assumes recruitment, mortality, and growth exactly balance; persistence is retained as the scoring benchmark the retention rule is defined against, not as a biological projection. Recreational catch remains incompletely measured.

The log-RMSE scores of Table 4 carry a floor caveat: structural trajectories absorb at the numerical floor $\varepsilon_{\log} = 10^{-3}$ kt rather than at zero (the trajectory code clips the state to $[\varepsilon_{\log}, 10^6]$ kt), and the reported values use $\log \max(\hat{S}, \varepsilon_{\log})$; $\varepsilon_{\log}$ is part of the registered scoring configuration and is distinct from the process noise ε_t of Definition 2.1. The floor binds often — for M1 and M1b it binds on a majority of five-year origins on both specifications — and the per-origin counts are given in Section SI-4. The raw-RMSE column is the retention score. A constant-productivity map is misspecified for a series whose assessment attributes the crash to $M \approx 2.5$; the test is whether that misspecification still forecasts SSB better than persistence, and the scored comparison answers that for this ladder on these series. Sample sizes are small ($n = 33$ years on Specification A; rolling $n = 25$ at $h = 1$ and $n = 21$ at $h = 5$). They suffice to rank models and do not suffice to certify a small skill difference. The 2023 40% B_{MSY} LRP is the Specification B setting, and Section 3.3 is that review: the ladder was run on the xteNCAM series under a fresh specification-matching review, with the same non-retention verdict.

Reconstruction-level corroboration. An independent, assessment-level record sharpens the same boundary. Rose (2026) compares two surplus-production reconstructions of the stock spanning 1983–2023 — the Rose and Walters (2019) model and the DFO (2024a) assessment model — and documents that surplus production and stock growth stalled after 2015, with some years negative, the stall-point biomass being controversial between the two reconstructions and well below historical norms. The recovery-stall window of Specification B (train 1995–2012, test 2013–2024) overlaps that period (its 2016–2024 portion). A stock whose surplus production and stock growth stall together, with some years negative and the stall-point biomass below historical norms — the configuration Rose (2026) reports — is precisely the configuration a constant-productivity surplus law cannot reproduce, and the ladder's single-origin comparison on that window is consistent with it on the rolling score, which is what decides retention. On that single fixed origin M1 (254 kt) and especially M1b (178 kt) do read below frozen persistence (262 kt); the retention verdict is nonetheless negative, because it is decided on rolling-origin RMSE, where persistence is not beaten at either horizon. The fixed-window result is reported because selecting favourable episodes would tell a different story from rolling evaluation, and that contrast is itself part of the finding. The two reconstructions' disagreement at the stall point is moreover the assessment-level twin of the ladder's own non-uniqueness: M1b's apparent recovery-window skill is carried by a carrying capacity extrapolated far beyond the training range, exactly the extrapolation on which the two reconstructions diverge. Rose (2026) also documents the limit-reference-point trajectory — successive downward revisions to just over 0.3 Mt, then to about 0.25 Mt in 2025 — which is the norm-field drift that separates the two specifications evaluated here (884.6 kt versus 276 kt) and the reason their columns are never pooled.

Attribution and the prey modules. Rose (2026) infers from structural-equation models that the post-2015 production deficit is limited by capelin and amplified by harp seal predation, with fishing almost certainly not the sole cause. That attribution is consistent with the ladder's two negative findings, and it sharpens both. First, it matches the catch-treatment null: neither prescribed catch treatment repairs the collapse, which is what one would expect if the dominant missing term is not catch. Both treatments could omit relevant removals, and other structural errors could prevent either from helping, so the scored result is consistent with that attribution rather than establishing it. Second, it draws the identification boundary inside the prey results: capelin limitation can be a real production driver — the 1991 acoustic collapse is large (1985–90 median 3704 kt versus 1991–2022 median 174 kt) — while the corresponding forecast modules still fail the stated score. A driver's reality at the reconstruction level does not make it a usable forecast module at the surplus-production level: the two-regime and acoustic-index variants are weakly constrained by their short training windows and missing index years, and they fail the retention score. No parameter profile was computed for them, so their non-retention rests on prediction error rather than on a demonstrated identification failure.

Protocol status. The evaluation windows and the scoring core — the primary RMSE comparison and the persistence bar — were coded before the first scoring pass and applied unchanged. Two components of the retention rule were not: the 5% tie band and the comparator declarations for the non-nested rungs were formalised after the scores of Section 3 had been computed. Neither affects any outcome reported here; the tie band never decides a retention result, the smallest deficit being 17%. The design is therefore a fixed computational protocol, not a prospective registration, and the additional passes (annual landings, survey start, capelin, Specification B) were declared in the text as they were added. No dated pre-scoring protocol file exists for this study. Section SI-1 records the order in which each pass was run.

The results are specific to the objects tested. They do not extend to other estimators of the same model family, to closed-loop filtering schemes, or across the two assessment specifications, which are scored separately throughout. The delay evaluated here is a one-year information delay in the forecast start state, not a continuous-time retarded system.

On this evidence, within the scope of this estimator and ladder, modules that are not identified on the training data did not reduce forecast error below matched persistence at both horizons here. Weak identification is documented for the recovery-window Allee fits and is not established as the sole cause of non-retention. The retained forecast is persistence.

5. Conclusions

On locked NCAM M-shift SSB for 1983–2015, last-value persistence is more accurate than surplus production, catch-driven stock-flow, autoregressive residuals, delayed information, and prey-informed productivity. The crash is not a catch-drop event in this accounting. Extra structure did not recover the gap to persistence at both horizons, and the stale-start module enlarged it; the autoregressive module improved on its own declared comparator without closing the gap to persistence. The same retention outcome holds on the unpooled xteNCAM series, whose post-2015 stall period fails in the same way the reconstructions stall.

The paper reports a forecast comparison. It does not conclude that the stock is unsustainable, and it does not conclude that the evaluation framework is empirically confirmed on this stock.

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Declarations

Data availability

All input data, analysis scripts, and result files are archived in the public repository at https://github.com/MIKEAA2020/general-sustainability; the frozen model specification is archived with them. Primary spawning-stock-biomass series: DFO (2016), Table A2. Alternative assessment series and landings: Regular et al. (2025), Tables 17 and 1. Historical landings: Schijns et al. (2021). Capelin acoustic index: Murphy et al. (2025), on the Northwest Atlantic Fisheries Centre (2025) Zenodo record 10.5281/zenodo.17515115. All computations are deterministic: within a fixed interpreter and library stack, repeated executions reproduce every result file byte for byte, and an independent re-execution of the scoring runners verified all 29 recorded output checksums (29/29 pinned) and regenerated all 30 result files byte-identically (30/30; one registered file carries no pinned checksum and is covered by the regeneration comparison). Across environments the record is: the intervention runners and the OLS-based prediction runners regenerate their outputs byte for byte, while the four optimizer-based forecast runners (L-BFGS-B fits) reproduce every scored row at printed precision except the M1b rolling row on Specification B, whose Allee optimum is environment-sensitive at the ± 17 kt level ($h = 1$: 151.6 versus 153.2 kt; $h = 5$: 445.5 versus 462.5 kt, on Python 3.12/numpy 2.1.3/scipy 1.14.1 versus the recorded original environment) — consistent with that model's identification fragility ($\mathfrak{s} \to 0$, with K resting at the 500-kt multi-start initialiser), and immaterial to the retention verdict, which persistence wins at both horizons by margins far larger than the instability. The archive records the checksums of the original-environment outputs; the deterministic-in-environment claim and this sensitivity note together are the reproducibility statement. The two assessment specifications (NCAM and xteNCAM) were analysed throughout as separate, unpooled series. The uncertainty layer (Section 3.5) is produced by `batch 7 (audits of agent arena 1 paper rewrites)/campaign_e1_dm_uncertainty.py` — seeded and deterministic, with Diebold–Mariano (HAC) and Kunsch moving-block bootstrap on the archived per-origin forecast files — and its output is archived alongside it as `batch 7 (audits of agent arena 1 paper rewrites)/results/e1_dm_uncertainty.csv`; the persistence baseline is recomputed there on the identical origin sets from the registered series and asserted against the recorded origin-matched values.

```
python3 src/run_ladder.py && python3 src/run_xte.py
python3 src/run_capelin_regime.py && python3 src/run_capelin_index.py
python3 src/compare_catch.py && python3 src/make_figures.py
```

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Declaration of competing interest

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AI declaration

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